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Namma Space: A Robust End-to-End Indoor Spatial Reconstruction & Free-Roaming Engine

Commodity Smartphone Monocular Video to Real-Time 60 FPS WebGL Gaussian Splats and Obstacle-Aware Navmesh

Keywords: 3D Gaussian Splatting, Monocular SfM, Standard Operating Procedure, Obstacle Occupancy Grid, WebGL Walkthrough.

Abstract—Creating high-fidelity, interactive digital twins of indoor environments has historically required expensive LiDAR hardware, complex photogrammetry clusters, or heavy proprietary desktop software. We present *Namma Space*, an end-to-end framework transforming ordinary monocular smartphone video of residential indoor spaces into a zero-install, responsive web application with real-time first-person free-roaming. The system integrates: (1) a formalized Standard Operating Procedure (SOP) ensuring high-overlap, blur-free video capture; (2) a dual-engine Structure-from-Motion (SfM) backend benchmarked between COLMAP incremental bundle adjustment and GPU-accelerated tracking; (3) 3D Gaussian Splatting (3DGS) with 3rd-order spherical harmonics optimized under a calibrated thermal profile (<70% VRAM on laptop GPUs); (4) automatic RANSAC floor-plane extraction and a 5 cm 2D occupancy navigation grid; and (5) a lightweight WebGL/WebGPU web viewer running sustained at 60 FPS with WASD collision-aware roaming and semantic POI search. Evaluated on real-world multi-room apartment environments, Namma Space achieves 31.5 dB PSNR, 0.938 SSIM, and a sub-35 MB streaming footprint, delivering true democratized indoor spatial digital twins.

1. INTRODUCTION & PROBLEM SCOPE

Accurate digital recreation of residential and commercial indoor spaces is vital for virtual tourism, real estate, architectural inspection, and autonomous indoor robotics. However, existing solutions exhibit severe practical bottlenecks:

- **Hardware Barrier:** Commercial scanners (e.g., Matterport Pro3) require multi-thousand-dollar LiDAR units inaccessible to general consumers.
- **Rendering Latency:** Classic Neural Radiance Fields (NeRFs) require dense volumetric raymarching, limiting client inference to low framerates (<10 FPS) without dedicated cloud rendering.
- **Geometry Artifacts:** Traditional multi-view stereo (MVS) produces noisy polygonal meshes on low-texture drywall, glossy tabletops, and transparent glass.

Namma Space bridges this divide by providing an automated monocular capture-to-web pipeline. By pairing 3D Gaussian Splatting with a formalized capture protocol and an automated floor-alignment navigation layer, users can capture any indoor room on a standard smartphone and immediately explore it inside a lightweight browser environment.

2. STANDARD OPERATING PROCEDURE (SOP)

Reconstruction fidelity in multi-view vision is strictly bounded by the geometric baseline and photometric consistency of input imagery. We formalize a rigorous, field-tested 4-step capture SOP:

2.1 Environmental Preparation Protocol

1. **Lighting Equalization:** Turn on all overhead artificial lighting and keep window blinds diffuse. Avoid direct noon sunlight causing extreme high-dynamic-range (HDR) shadows.
2. **Motion Quarantine:** Remove moving objects, pets, and dynamic clutter. Ensure doors between connected rooms remain stationary.
3. **Camera Rig Setup:** Standard smartphone (iPhone/Android) recording at 1080p or 4K. Lock Auto-Exposure and Auto-Focus (AE/AF lock) to eliminate exposure flicker and focal breathing between frames.

2.2 Spatial Trajectory & Motion Execution

Camera motion follows a dual-loop geometric trajectory:

- **Perimeter Loop:** The operator walks slowly along the room walls (walking velocity $v \le 0.4 ext{ m/s}$), holding the phone at chest level ($h pprox 1.4 ext{–}1.6 ext{ m}$), angled $ pprox 15^\circ$ inward toward the center.
- **Interior Cross-Sweep:** A central diagonal or figure-8 pass capturing table surfaces, ceiling corners, and occluded furniture under-edges.
- **Overlap Criterion:** Maintain $\ge 75\%$ forward inter-frame overlap and $\ge 60\%$ cross-track baseline overlap.
- **Loop Closure:** The trajectory strictly terminates at the starting physical waypoint, establishing closed loop-closure constraints in bundle adjustment.

2.3 Blur Rejection Filter

To eliminate walking-induced motion blur, keyframe extraction executes Laplacian variance filtering:

$$ ext{Var}(\Delta I) = rac{1}{N} \sum_{x,y} \left(abla^2 I(x,y) - ar{ abla}^2 I ight)^2 \ge au_{ ext{blur}}$$

Frames exhibiting $ ext{Var}(\Delta I) < 15.0$ are dynamically discarded, guaranteeing pristine sharp feature descriptors.

3. SYSTEM ARCHITECTURE & PIPELINE

The Namma Space automated pipeline executes across 7 decoupled, fault-tolerant stages:

End-to-End Processing Flow:
Raw Video $ o$ [1. Keyframe Sampling & Blur Filter] $ o$ [2. Pose Backend (COLMAP / Postshot)] $ o$ [3. RANSAC Floor Alignment] $ o$ [4. 3DGS Optimization] $ o$ [5. 2D Occupancy Grid Slicing] $ o$ [6. Binary Compression] $ o$ [7. Web Viewer].

3.1 Pose Estimation & Benchmarks

Accurate extrinsic ($[R_i | t_i]$) and intrinsic (K) camera parameters are solved by minimizing multi-view reprojection error:

$$\min_{\{R_i, t_i, X_j\}} \sum_{i,j} ho\left(\| x_{ij} - \pi(K, R_i X_j + t_i) \|^2 ight)$$

Namma Space supports both incremental CPU bundle adjustment (COLMAP) and high-speed GPU Structure-from-Motion (via Postshot). Benchmark comparisons on an RTX 4060 laptop are summarized in Table 1.

Pose Backend	Runtime	Tracked	Residual	Memory
COLMAP (12 P-threads)	32.3 s	100.0%	1.16 px	2.8 GB RAM
Postshot GPU SfM	**18.4 s**	**100.0%**	**0.84 px**	**3.5 GB VRAM**
VGGT Windowed	68.9 s	100.0%	0.35 px	3.8 GB VRAM

Table 1: Pose backend benchmark on 30-frame indoor dataset.

3.2 Floor Plane Alignment

SfM reconstructions yield arbitrary scale and coordinate orientations. Namma Space automatically fits a floor plane using RANSAC on the lower 40% height quartile of sparse points:

$$ax + by + cz + d = 0, \quad \hat{n}_{ ext{floor}} = [a, b, c]^T$$

A closed-form Rodrigues rotation $R_{ ext{align}}$ maps $\hat{n}_{ ext{floor}}$ directly to the world vertical vector $[0, 1, 0]^T$ (Y-up), shifting the floor plane to $Y = 0.0 ext{ m}$. This guarantees seamless camera eye-height ($1.65 ext{ m}$) across all viewers.

4. 3D GAUSSIAN SPLATTING OPTIMIZATION

Scene radiance is modeled by a continuous set of N 3D Gaussians $G(x) = \exp\left(-rac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu) ight)$, parameterized by:

- Position centroid $\mu \in \mathbb{R}^3$
- Covariance matrix $\Sigma = R S S^T R^T$ decomposed into scaling vector $S \in \mathbb{R}^3$ and unit quaternion rotation $q \in \mathbb{R}^4$
- Alpha opacity $lpha \in [0, 1]$ via sigmoid activation
- View-dependent color via 3rd-order Spherical Harmonics ($l_{ ext{max}} = 3$, 16 coefficients per color channel).

4.1 The "Golden Balance" Thermal Profile

Running unconstrained 4K 3DGS on laptop GPUs easily triggers thermal throttling and VRAM micro-paging ($>85\%$ VRAM capacity). Namma Space implements a calibrated balanced training profile:

- **Clamped HD Resolution:** Input images are clamped isotropically such that $\max(W, H) \le 1600 ext{ px}$. This reduces per-iteration rasterization overhead by 31% while preserving sub-pixel splat fidelity.
- **CPU Dataset Buffering:** Dataset camera buffers are held in system RAM (`--data_device cpu`), reserving GPU memory strictly for active Gaussians and backward rasterizer tiles ($3.5 ext{–}5.2 ext{ GB}$ VRAM).
- **10,000 Iteration Convergence:** Optimization reaches full 3rd-order spherical harmonic convergence and densification in $pprox 5 ext{ minutes}$ with $31.5 ext{ dB}$ PSNR, avoiding the diminishing returns of 30,000 steps.
- **Safety Multi-Checkpoints:** Models are automatically checkpointed at 7k, 10k, and 15k steps.

5. NAVIGATION & OCCUPANCY MAPPING

To enable realistic physical roaming without passing through walls, a 2D floor-aligned occupancy grid is generated at a fine $5 ext{ cm}$ cell resolution:

1. **Obstacle Slicing:** Splat centroids within the height slice $Y \in [0.20 ext{ m}, 1.80 ext{ m}]$ (above baseboards and below ceiling fixtures) are projected onto the horizontal (X, Z) ground plane.
2. **Morphological Filtering:** Morphological opening ($3 imes 3$ kernel) eliminates isolated point noise.
3. **Agent Radius Inflation:** Walls and obstacles are dilated by an agent radius $r_{ ext{agent}} = 0.30 ext{ m}$ to prevent camera clipping during first-person roaming.
4. **A* Pathfinding:** Waypoint graphs and shortest walkable paths are computed on demand.

6. WEB VIEWER & FREE-ROAMING ENGINE

The client application is built with React 18, Three.js, and `@react-three/drei` Splat rendering, compiled via Vite into an ultra-fast production bundle ($<200 ext{ KB}$ JS):

- **Direct Splat Rendering:** The 3D scene loads directly from a pre-compressed 32-byte binary format (`.splat`), rendering millions of Gaussians in real time via WebGL/WebGPU.
- **WASD First-Person Roam:** Incorporates click-to-lock mouse look, smooth WASD translation, `Shift` sprint ($2.5 imes$), and `Space`/`C` vertical inspection.
- **Dual Inspection Modes:** One-click toggle between 🕹️ WASD Roam and 🔄 Orbit View for macro-spatial examination.
- **Orientation Correction:** In-viewer 90° pitch/roll alignment buttons allowing users to correct phone orientation instantly.
- **POI Search:** Real-time fuzzy POI search powered by Fuse.js with camera fly-to interpolation.

7. EXPERIMENTAL RESULTS & EVALUATION

Evaluation across multiple domestic apartment environments (Ex1, Ex2, Ex3, Ex4) demonstrates state-of-the-art visual quality and real-time client performance (Table 2).

Metric	Target Criteria	Namma Space	Status
PSNR (Reconstruction)	> 28.0 dB	**31.52 dB**	Exceeded
SSIM (Fidelity)	> 0.880	**0.938**	Exceeded
Web FPS (Desktop)	≥ 30 FPS	**60 FPS**	Exceeded
Web FPS (Mobile)	≥ 25 FPS	**48 FPS**	Exceeded
Model Payload Size	< 50 MB	**27.4 MB**	Exceeded
Training Runtime	< 15 mins	**5.2 mins**	Exceeded

Table 2: Performance metrics against competition evaluation benchmarks.

8. CONCLUSION

Namma Space proves that consumer smartphones, paired with principled Standard Operating Procedures and modern Gaussian Splatting, can deliver photoreal indoor digital twins that run effortlessly inside standard web browsers. The framework sets a new standard for accessible, high-performance spatial computing.